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Short communication

Development of aerobic methane oxidation, denitrification coupled to methanogenesis (AMODM) in a microaerophilic expanded granular sludge blanket biofilm reactor

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ABSTRACT

The issue of enhancing nitrogen removal and managing dissolved methane emission in anaerobic treatment systems is a major bottleneck in its wider application to treat high-strength organic wastewater with nitrate. Herein, a novel aerobic methane oxidation, denitrification coupled to methanogenesis (AMODM) process was developed in a glucose-fed microaerobic expanded granular sludge blanket biofilm reactor (EGSBBR) through *in-situ* utilization of produced methane for nitrogen removal. The 162-day operation demonstrated that long-term treatment performance under the decreased COD/NO_3^--N (C/N) ratio from 66.7 to 10 and the optimal C/N ratio for completing AMODM was found to be 16.7. Microbial community analysis further evidenced that *Methanothrix* as key methanogen predominated in the sludge bed, while *Methlogaea* as aerobic methane oxidizer was mainly detected in the packing bed of the hybrid system. Meanwhile, some facultative heterotrophic and dissimilated nitrate-reduction (DNRA) genera also co-existed. The profiling of key functional genes further proved concurrent occurrence of methanogenesis, aerobic methane oxidation and denitrification. Furthermore, possible microbial mechanism on AMODM process was elucidated from the prospective of targeted species interaction within the reactor. This research provides a robust and environment-friendly alternative process treating nitrate-containing organic wastewater towards efficient nitrogen removal, low resource consumption, bioenergy recovery and greenhouse gas reduction.

1. Introduction

Anaerobic digestion (AD) known as an energy-positive biotechnology has been extensively applied for high-strength organic wastewater treatment (Rajeshwari et al., 2000) due to its attractive benefits of no oxygen supply and biogas (gaseous methane; G-CH_4) recovery. However, amounts of dissolved methane (D-CH_4) still remains in AD effluent due to its supersaturation (up to 22–28 g CH_4/m^3; 25–35 °C) (Kim et al., 2010; Souza et al., 2011) and fails to separate and reuse from the liquid phase. Even worse, D-CH_4 may escape into the atmosphere and cause a strong greenhouse effect (IPCC, 2013) if it subjects to stripping like aeration (Daelman et al., 2012). Another major problem in the widespread application of the AD process is the limitation in removing nitrogen, especially for nitrogen-containing carbonaceous wastewater. To overcome it, one effective strategy is the sequential treatment of anaerobic followed by aerobic treatment (Akizuki and Toda, 2018). As such, nitrogen and organics can be removed by simultaneous denitrification and methanogenesis (SDM) (Andalib et al., 2011) through recycling nitrifying liquid back to the front AD reactor. Nevertheless, its nitrate removal is largely affected by the availability of organic matters, which are competed between methanogens and denitrifiers. Moreover, D-CH_4 emission is still unsolved in SDM process.

Recently, aerobic methane oxidation (AMO) as a newly important pathway in global carbon and nitrogen cycles has been firstly discovered in wastewater treatment systems (Ho et al., 2013). Particularly, AMO means that methane is capable of being oxidized into methanol and other oxidative products, thus providing alternative carbon sources for enhanced denitrification, namely aerobic methane oxidation coupled to

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denitrification (AME-D) (Zhu et al., 2016). Based on microbial findings and potential possibilities, CH_4 *in-situ* generated from the AD process is expected to be directly utilized by AMO microorganisms to intensify nitrogen removal by means of the integration of SMD and AME-D into a single methanogenic reactor under microaerophilic condition. Therefore, synchronous removals of organics, nitrogen and D-CH_4 with partial recovery of G-CH_4 can be realized via aerobic methane oxidation, denitrification coupled to methanogenesis (AMODM). As far as our knowledge, however, it has not been exploited yet so far.

The main purpose of the present study is to investigate the feasibility of AMODM in an expanded granular sludge blanket biofilm reactor (EGSBBR) to treat nitrate-containing organic wastewater and to unravel possible microbial mechanisms of AMODM based on the identification of key functional bacteria and genes using 16S rDNA sequencing technique.

2. Material and methods

2.1. Reactor setup

A lab-scale continuous up-flow EGSBBR system was set up in Fig. S1. The effective volume of the whole system was 17 L (1.9 m height and 9.0 cm inner diameter) was composed of main body (9 L) and triphase separator (8 L). The main body included an expanded granular sludge bed (bottom) and a packing bed (upper) filled with 25% (v/v) sponge cubes (8 × 8 × 8 mm) as media. The separator was fixed at the reactor top to separate treated wastewater, precipitate the sludge and collect the biogas. The biogas flowrate was measured by an electronic gas flowmeter and treated effluent was discharged from the weir located at the separator. The whole system was controlled at 35 ± 1°C by a heater for promoting mesophilic methanogenesis.

2.2. Operation procedure

The reactor was inoculated with the mixture at a 2:1 vol ratio of thickener sludge (MLSS: 15.76 g/L) and aerobic activated sludge (MLSS: 7.47 g/L) from a municipal wastewater treatment plant in Jinzhong city, China. COD and NO_3^--N were added in the synthetic wastewater in the form of glucose, sodium nitrate, respectively. The alkalinity and phosphorus source were provided by $NaHCO_3$ and KH_2PO_4 and 1 mL/L trace element solution (5 g EDTA, 2 g $FeCl_3{\cdot}4H_2O$ 0.2 g $CoCl_2{\cdot}6H2O$, 0.5 g $MnCl_2{\cdot}4H_2O$, 0.03 mg $CuCl_2{\cdot}2H_2O$; 0.05 g $ZnCl_2$, 0.1 g Na_2SeO_3 , 0.05 g $NiCl_2{\cdot}6H_2O$, 0.09 g Na_2MoO_4, 0.05 g H_3BO_4 per liter).

To accelerate methanogenesis during anaerobic digestion, the microaerobic pretreatment as a preferred method was used (Montalvo et al., 2006). In this study, low dissolved oxygen (DO) and oxidation-reduction potential (ORP) under the liquid level 20 cm of the triphase separator were maintained at 0.22 ± 0.02 mg/L and −340 ± 30 mv, respectively to ensure the co-cultivation of methanogens, denitrifiers and aerobic methanotrophs. The pre-aerated feeding was pumped into the bottom of the up-flow reactor at ascending liquid velocity of 1.2 m/h by a peristaltic pump with 100% internal recycling ratio. The hydraulic retention time (HRT) was set to 2 d and no sludge was discharged resulting in the mixed liquid volatile suspended solid (MLVSS) increased from initially 8.2 g/L to finally 15.0 g/L during the experiment. During the start-up stage, the reactor was initially fed with high-strength organic simulate wastewater only with the influent COD of 2000 mg/L and the influent pH was maintained at 7.2 ± 0.1 to enrich methane-producing bacteria. After the 46-day start-up period, lots of biogas bubbles were visibly observed, denoting the completion of methanogenesis. And then a certain amount of NO_3^--N began to be added into the feed. To explore the feasibility of AMODM, five different operations at C/N ratio from 66.7 to 10 by adjusting influent NO_3^--N concentration were carried out during the 162-day experiment, while COD was kept always constant, shown in Table 1.

Table 1
Operation conditions during the experiment.

Phase	Day	COD (mg/L)	NO_3^--N (mg/L)	C/N ratio
I	1–40	2000	30	66.7
II	41–64	2000	60	33.3
III	65–88	2000	90	22.2
IV	89–123	2000	120	16.7
V	124–162	2000	200	10

2.3. High-throughput sequencing

The samples of activated sludge in the sludge bed and biofilm in the packing bed was collected and stored in the refrigerator at −20 °C on Day 116 (Phase IV). The microbial community were analyzed by the Illumina high-throughput sequencing platform based on 16 S rDNA by Sangon Botech Co., Ltd (Shanghai, China) according to our previous study (Zhou et al., 2018). Key functional genes were compared and estimated by the Kyoto encyclopedia of genes and genomes (KEGG) database (https://www.kegg.jp) (Zhou et al., 2020).

2.4. Chemical analysis

NH_4^+-N, NO_2^--N, NO_3^--N and COD concentrations were measured by HACH Lange cuvette kits. TN was the sum of NH_4^+-N and NO_x^--N. pH, DO and ORP were monitored by a portable analysis meter (WTW 3420, Germany). The CH_4 and CO_2 concentrations of the gas samples over the headspace were determined by a gas chromatograph (GC-4000A, China) equipped with flame ionization (FID) detector and thermal conductivity (TCD) detector, respectively.

3. Results and discussion

3.1. Reactor performance

To rapidly establish AMODM in a methanogenic system, the influent C/N ratio as a crucial process parameter was mainly investigated. Fig. 1 shows the overall time-course performance of the reactor under different influent C/N ratios. As shown in Fig. 1a, steadily low effluent COD (39.5 ± 12.3 mg/L) with the corresponding COD removal efficiency of 98.1 ± 0.7% was obtained in the whole experiment. The biogas production and influent and effluent pH are present in Fig. 1b. From the chart, methane yield as high as 608 mLCH_4/d or 35.7 mL CH_4/(L·d) was achieved in the PhaseI, suggesting successful build-up of methanation at the influent organic loading rate (OLR) of 1.0 kg COD/(m^3·d). Subsequently, only a bit decrease in methane production was observed when the nitrate was increasingly added in PhasesII-III. Up to 120 mg/L nitrate in Phase IV, the generation of methane lowered to 403 mLCH_4/d or 23.7 mL CH_4/(L·d), which was nearly 2/3 of its maximum production. The rise of the effluent pH in Fig. 1b also implied the satisfactory methanogenesis performance owing to evident accumulation of alkalinity during the methanation metabolism. Afterward, the CH_4 production sharply declined to below 200 mL/d or in Phase V by decreasing C/N ratio to 10. As the nitrate was added, more organics would be transformed by denitrification into CO_2 instead of CH_4, resulting in the decline of methane yield.

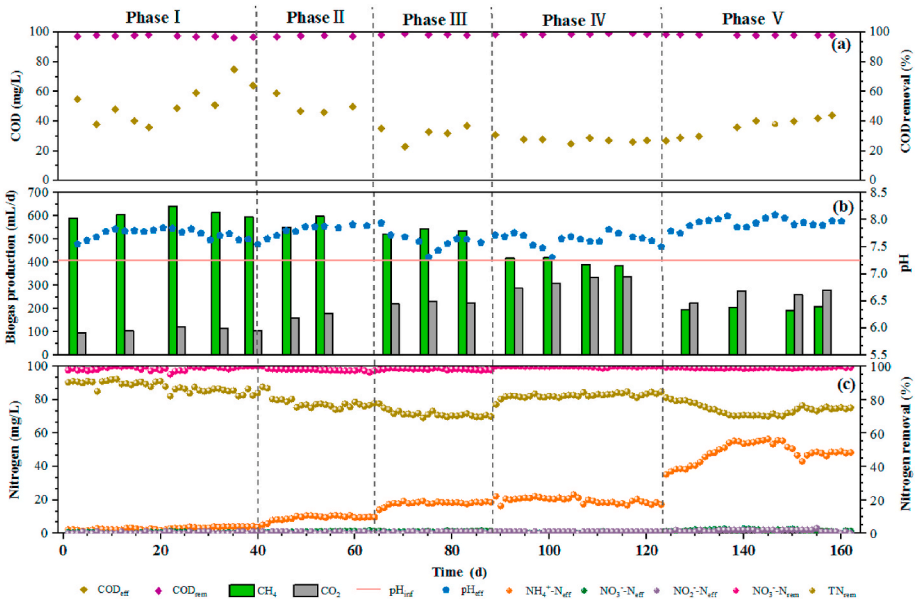


Fig. 1. Reactor performance during the operation: (a) Effluent COD and COD removal; (b) Biogas production and influent and effluent pH; (c) Effluent ammonia-, nitrite- and nitrate-nitrogen and nitrogen removal.

As for nitrate removal, as high as 98.9% NO_3^--N removal rate was accomplished in the whole operation in Fig. 1c. In comparison with stably complete NO_3^--N removal, however, average TN removal underwent a gradual decrease from 87.7% to 71.2% as NO_3^--N increasing in the first three stages. But surprisingly, the nitrogen removal capability in Phase IV stopped falling but improved to 82.7% on the average instead. Taking decreased CH_4 and increased CO_2 production (in Fig. 1b) into consideration, methane oxidation might have play a significant role in unexpected improvement of TN removal. Under the microaerobic condition, methane was firstly oxidized into methanol by aerobic methane-oxidizing bacteria, which was thrived in stable methane-producing and nitrate-rich environment and then reduced nitrate to nitrogen gas utilizing oxidation products of methane as alternative carbon source for denitrification.

Nevertheless, it was noted that TN removal efficiency significantly decreased and then stabilized up to nearly 75% if influent NO_3^--N further increased to 200 mg/L in Fig. 1c. In terms of sharp decline in methane generation and TN removal, methanogenesis was greatly inhibited and even deteriorated due to the low C/N ratio. Andalib et al. (2011) also found that methanogenesis would lose the competition with denitrification if the C/N ratio dropped to below 10 in the SDM process. Likewise, the denitrification gradually dominated in nitrogen removal in ours study, as more nitrates were served as electronic acceptor for denitrification. That was why TN removal was recovered somewhat in the late period of Phase V. During the operation, effluent TN was almost present in the form of NH_4^+-N, indicating dissimilatory nitrate reduction to ammonia (DNRA) possibly involved in the nitrogen conversion. Andalib (Andalib et al., 2011) also reported the occurrence of DNRA during organic wastewater treatment at the presence of methane and nitrate. Overall, AMODM was successfully accomplished through adjusting C/N ratio via the coupling of methanogenesis, methane oxidation and denitrification in the single methanogenic reactor.

3.2. Analysis of functional bacteria and key genes

High-throughput sequencing was conducted to further identify the microbial community and associated genes on Day 116 in Phase IV. Fig. 2a depicts the major microbial community (abundance >0.5%) at the genus level in both the sludge and biofilm. According to the analysis, heterotrophic microorganisms including anaerobic/facultative bacteria (*Saccharibacteria_genera_incertae_sedis*, *Smithella*, *Trichococcus*, *Thermogutta*, *Ornatiline*, *Longilinea*) participating in fermentation and acidification of glucose and denitrifiers (*Rhizobium*, *Rhodobacter*, *Aridibacter*, *Hyphomicrobium*) were predominated, accounting for 42.2% of the total consortium for activated sludge sample and 39.8% for biofilm sample. *Methanothrix* accounting for 6.21% was widely discovered in the sludge bed of the reactor. As a common member of acetoclastic methanogens (Zinder et al., 1984), *Methanothrix* were suitable to be grown in the acetic acid environment. Moreover, lots of fermentative bacteria rapidly decomposed glucose into volatile fatty acid (VFA), ensuring the enrichment of methanogens. In contrast to methanogens, *Methlogaea* (1.53%) belonging to aerobic methanotrophic bacteria (Tarlera, 2015) were mainly recognized in the biofilm sample. The setting of the packing bed filled with small-sized sponge could delay the ascending velocity of the methane gas. Thus *in-situ* capture of the methane might be effectively completed by aerobic methane oxidizers to promote further conversion of the methane. Meanwhile, products of aerobic methane oxidation could be used as favorable carbon sources for nitrate denitrification. The large discovery of *Methanothrix* also confirmed this point. Detection of some genera (*Desulfobulbus*, *Planctmicrobium*) in both activated sludge and biofilm samples suggested the reduction of nitrate to ammonia (DNRA) pathway (Sousa et al., 2017) due to the presence of enough

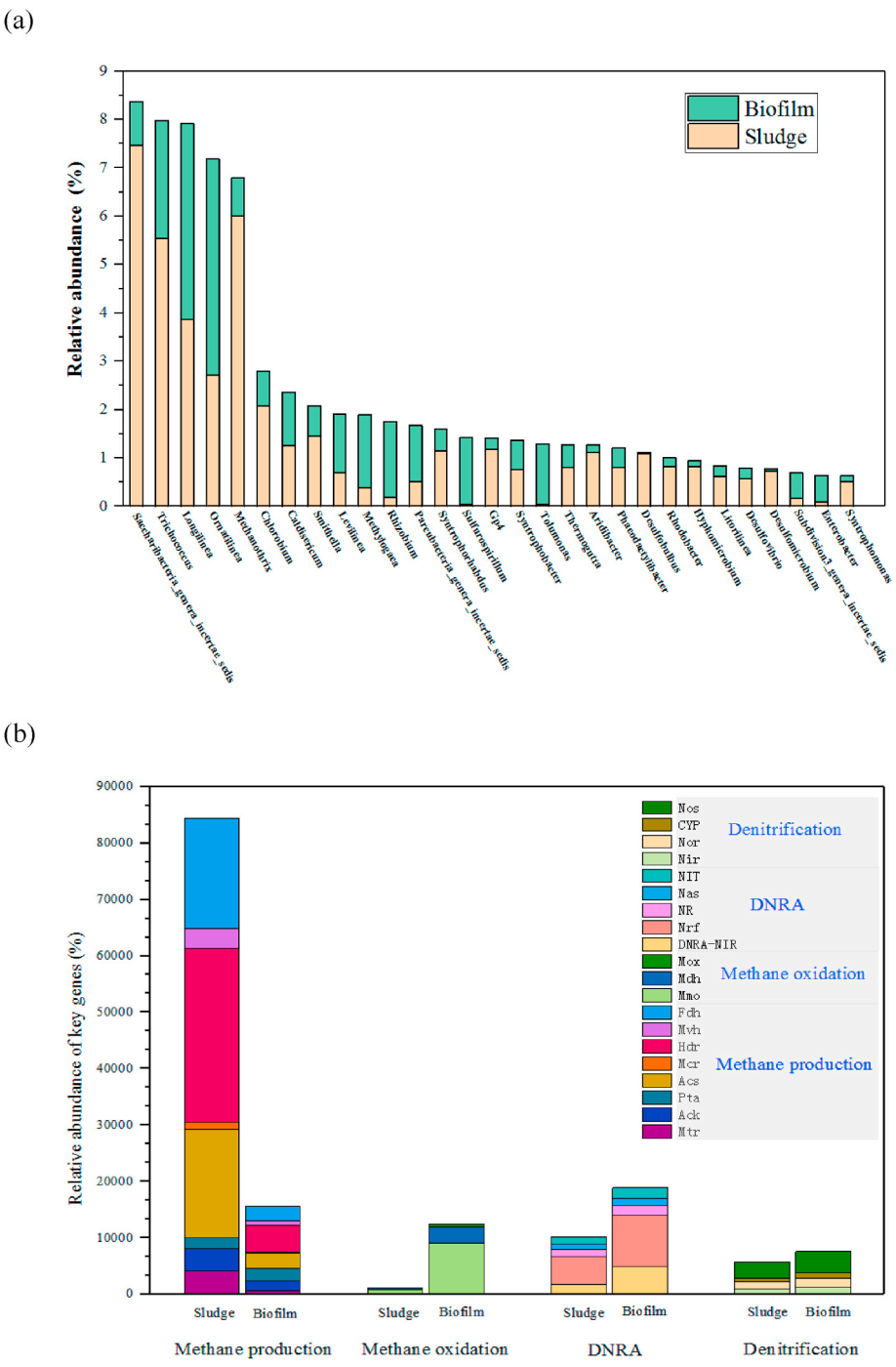


Fig. 2. Microbial analysis: (a) Bacterial community compositions at the genus level; (b) Key functional genes.

dissolved methane occurred. This directly led to high accumulation of ammonia in the final effluent in Figs. 1c and 2.

Fig. 2b illustrates the abundance of key functional genes involved in microbial metabolisms. Apparently, there was a positive correlation between the profiling of the functional genes and the composition of microbial consortium. A high relative abundance of key functional genes (Acs, Hdr, Fdh, Mvh) enabled to execute methane production (Sakai et al., 2011). Identification of key functional genes (Mmo, Ndh) verified the functional expression of aerobic methane oxidation (Dumont and Murrell, 2005). Besides, some major genes (Nir, Nor, CYP, Nos) and genes (Nrf, DNRA-NIR, Nit, Nis) were found to be associated with denitrification and DNRA, respectively (Takaya, 2002). These microbial results could highly imply that the co-recognition of core genera and key genes prompted the AMODM process for efficient removal of carbon and nitrogen as well as methane recovery in one-stage microaerophilic reactor.

3.3. Microbial mechanism

To deeply understand fundamental of the SMODM, the wastewater samples on day 118 (Phase IV) from eight-height sampling ports (influent, S_0–S_5 and effluent) in Fig.S1 were analyzed for the profile of COD and nitrogen along the reactor, as shown in Fig.S2. Obviously, the overall changing trends in COD and NO_3^--N were closely associated with plug-flow pattern of the EGSBBR reactor. Firstly, COD and NO_3^--N concentrations were fast diluted by internal recirculation. Along the reactor

height, both indicators further went down, particularly in the sludge bed at the reactor bottom. Based on microbial community analysis in section 3.2, facultative fermentation and methanogenesis were responsible for organic removal, while nitrate was mainly reduced to nitrite and even ammonia there attributed by DNRA. At port S_3, it was observed that NO_2^--N and NH_4^+-N were accumulated to 24 mg/L and 8.5 mg/L, respectively, resulting in no obvious TN removal. As the flow upward, however, both nitrate and nitrite decreased in stepwise in the packing bed along with consistent decline of COD. As the reduction of NO_x^--N was performed by denitrification using products of aerobic methane oxidizers driven by gene Mmo in Fig. 2, AME-D distinctly took place. Meanwhile, NH_4^+-N slightly increased to 21.5 mg/L in the final effluent, indicating the co-existence of DNRA and denitrification. According to mass balance, 73.1% COD were removed in the sludge bed, while the packing bed was a major zone for nitrogen removal, occupying 61.4% of TN removal of the whole system. Owing to the SMODM, complete COD and nitrate removal as well as over 80% TN removal could be achieved.

From the comprehensive results from treatment efficiency, microbial findings and removal characteristics of the reactor, possible biochemical reactions and standard Gibbs free energies ($\Delta G^{0,}$) involved in SMODM were proposed in the microaerophilic EGSBBR in reference to (Andalib et al., 2011; Conrad, R. 1999; Zhu et al., 2016):

Reaction equations $\Delta G^{0,}$ (kJ/mol)

In the sludge bed:

Aerobic oxidation	$C_6H_{12}O_6+6O_2 \xrightarrow{OHB} 6CO_2+6H_2O$	−2880
Fermentation	$C_6H_{12}O_6 \xrightarrow{FHB} VFA \xrightarrow{FHB} 3CH_3COOH$	−311.2
Methanogenesis	$CH_3COOH \xrightarrow{MPB} CH_4+CO_2$	−35.6
DNRA	$C_6H_{12}O_6+12NO_3^- \xrightarrow{DNRA} 12NO_2^-+6H_2O+6CO_2$	−1767
DNRA	$C_6H_{12}O_6+4NO_2^-+8H^+ \xrightarrow{DNRA} 4NH_4^++2H_2O+6CO_2$	−1713

In the packing bed:

AMO	$2CH_4+O_2 \xrightarrow{MOB} 2CH_3OH$	−62.3
AMO	$2CH_3OH+3O_2 \xrightarrow{MOB} \ldots \xrightarrow{MOB} 2CO_2+4H_2O$	−115.5
Denitrification	$5CH_3OH+6NO_3^-+6H^+ \xrightarrow{DHB} 3N_2+13H_2O+5CO_2$	−109
DNRA	$C_6H_{12}O_6+4NO_2^-+8H^+ \xrightarrow{DNRA} 4NH_4^++2H_2O+6CO_2$	−1713

According to above equations, the possible removal mechanism of the AMODM was illustrated as Fig. 3. At the bottom of the reactor, microaerobic condition promoted the enrichment of facultative (FHB) and even oxic (OHB) heterotrophic bacteria suitably grown in the sludge bed with high biomass, which shortly transformed glucose into low-molecular short-chain VFAs and then to acetic acids. After that, methane was likely to be produced by acetotrophic methane-producing bacteria (MPB) through decarboxylation of acetate. In this study, rather low DO (about 0.2 mg/L) not only had no adverse effects on the growth activity of MPB, but promoting methanogenes because of the rapid accumulation of VFAs through microaerobic fermentation of glucose (Fu et al., 2020; Montalvo et al., 2016). As the pre-aerated liquid went upward in the reactor, the produced methane gas was probable to be captured and fixation by the biocarriers in packing bed, offering desirable attached-growing environment for mass proliferation of *Methlogaea* as aerobic methane oxidizers (AMOs), confirmed by microbial analysis in Fig. 2a. Moreover, the prolonged contact time of biomass with methane also favored the growth of the AMOs owing to gas mass transfer limitation in the packing bed. In this case, both oxygen and methane was consumed by AMOs, which aerobically transformed methane into methanol and provided alternative carbon sources for denitrifiers to reduce nitrate. Methanol also could be further oxidized by AMO to methanal (HCHO), formic acid (HCOOH) and even carbon dioxide (CO_2) based on complete genetic pathways of aerobic methane oxidation depicted in Fig. 2b. Furthermore, it was observed that DNRA was a main pathway for nitrate reduction in both sludge and biofilm. As reported in

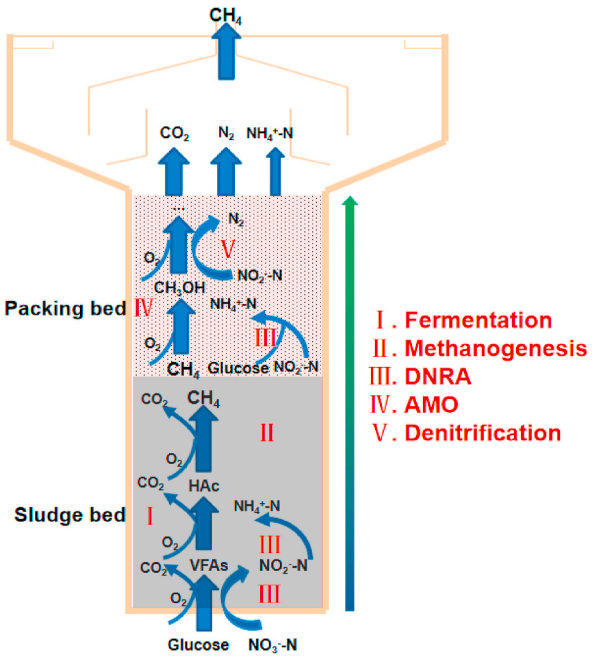


Fig. 3. Possible mechanism of the AMODM process.

previous studies (Andalib et al., 2011; Ruiz et al., 2006), it was easier for DNRA to occur at relatively high C/N ratio and nitrate-limiting condition, although denitrification was a competitor with DNRA for nitrate. In this study, as high as C/N (>10) and a relatively limited nitrate (<200 mg/L) would favor the rapid growth of DNRA bacteria, which outcompeted denitrifiers in sludge bed under strongly low redox conditions (ORP < −300 mv). Additionally, the occurrence of denitrification or DNRA also depended on the nature of carbon source (Akunn et al., 1993). Especially when the glucose as carbon source was available, it was likely to reduce nitrate via DNRA pathway. While VFAs or alcohols were used as carbon source, denitrification favorably occurred. These could also well explain the accumulation of ammonia in the effluent and the presence of functional genes of DNRA. As a result of large consumption of glucose, denitrification driven by aerobic methane oxidation (AME-D) was enhanced in methane-rich packing bed, as methanol replacing glucose was served as carbon source. On account of effective operating strategy and desirable reactor configuration, it could be strongly believed that the co-existence and collaboration of methanogens, methanotrophs and denitrifiers was decisive to AMODM in a microaerophilic single system.

4. Conclusion

In this study, AMODM process was successfully developed in a lab-scale EGSBBR to make the most of the dissolved methane *in-situ* produced in the single methanogenic system to enhance nitrogen removal. The system could averagely obtain 98.8% COD and 82.7% TN removal efficiency as well as reduce 1/3 of the methane emission at the optimal C/N ratio of 16.7 using AMODM, which also recovered approximately 23.7 mL/(L·d) CH_4. High-throughput sequencing analysis further verified key functional bacteria and genes involved in AMODM. The reaction mechanism of AMODM was elucidated based on the interaction of methanogenic, methanotrophic, denitrifying and DNRA bacteria selectively enriched in sludge and biofilm phases of the EGSBBR.

Credit author statement

Xin Zhou: Conceptualization, Methodology, Writing - original draft, Writing - review & editing. **Gonglei Wang:** Writing - original draft.

Daling Ge: Reactor operation, Data curation. **Zeyang Yin:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2020.111280.

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